

Social Science Inquiry II

Week 8: Statistical power, effect sizes, sample size

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Today's questions

- What is power?
- What drives it?
- How do we justify sample size?

Learning goals

- Define power and connect it to false negatives.
- Identify the main ingredients that determine power.
- Explain what makes a power analysis believable (or not).
- Write a defensible sample-size justification.

Power: signal vs. noise

Coppock (2022)

- **Power** is the ability to distinguish signal from noise. Formally: the probability of rejecting the null hypothesis when the alternative is true (i.e., detecting a real effect).
- Signal: the treatment effect (or relationship) we care about.
- Noise: how much outcomes vary for other reasons.
- High noise + small effects $\Rightarrow$ hard to detect.

Errors you care about

	Reject H_0	Fail to reject H_0
H_0 is true	Type I error (α)	Correct
H_A is true	Correct (power = $1 - \beta$)	Type II error (β)

- Low power means a high chance of missing real effects.

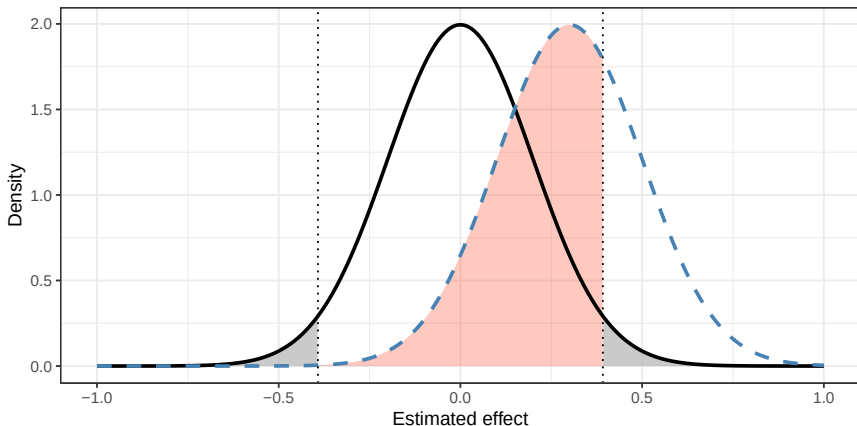
Normal critical values and z-quantiles

- Under a normal approximation, test cutoffs are measured in “standard error units.”
- Notation: z_p is the p th quantile of $N(0, 1)$ (so $\Pr(Z \leq z_p) = p$ for $Z \sim N(0, 1)$).
- Common values: $z_{0.975} \approx 1.96$ (two-sided $\alpha = 0.05$) and $z_{0.80} \approx 0.84$ (power = 0.80).

Visualizing α , β , and power

Sampling distributions under H_0 and H_A

Two-group design; n per group = 50, sd = 1, delta = 0.3, alpha = 0.05, power = 0.32



- Gray tails under H_0 : Type I error (α).
- Red region under H_A : Type II error (β). Power = $1 - \beta$.

Why we need power analysis

- If an effect exists, will our study design be able to detect it?
- Underpowered studies are costly and uninformative.
- Power analysis forces us to make our assumptions explicit:
 - How big is the effect we care about?
 - How noisy is the outcome?
 - What sample size and design can we actually run?

Quick check

If power = 0.30, what does that mean?

The three ingredients of power

- **Effect size:** how big is the signal?
- **Noise:** how variable is the outcome?
- **Sample size:** how much information do we collect?
- Design choices also matter (assignment ratio, clustering, blocking) and estimation choices (estimator, covariate adjustment).

Effect size isn't "what you hope"

- The effect size used in power analysis should be tied to a claim.
- Often, the key question is: **what is the smallest effect size of interest?**
- If we don't define what effects matter, "powered" doesn't mean much.

Noise comes from measurement and design

- Outcome variability (heterogeneity in the population).
- Measurement error and survey noise.
- Timing (when outcomes are measured).
- Noncompliance and attrition.
- Dependence (clusters, repeated observations).

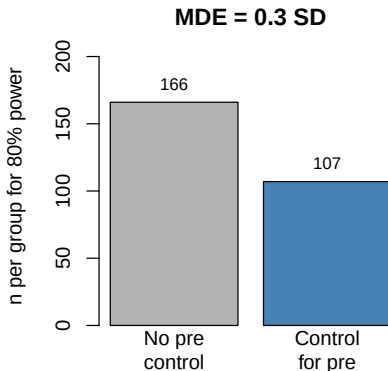
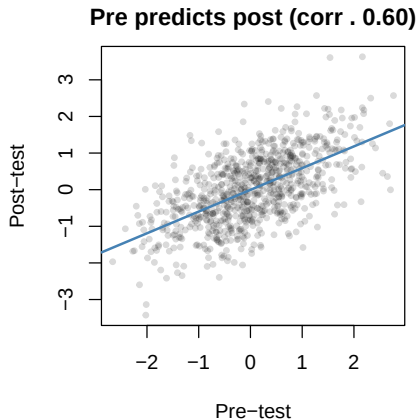
N is not the only lever

- Increasing N increases power, but:
- You can sometimes *increase precision* without increasing N :
 - better measurement (reduce noise)
 - better design (blocking/stratification)
 - better analysis plan (pre-specified covariates)

Example: pre-test covariates can boost power

Offer-Westort et al. (2026)

- If a pre-test strongly predicts the post-test, controlling for it reduces residual noise.
- Less noise $\Rightarrow$ smaller n for the same effect size of interest and target power.



Applications and tradeoffs

1. For each case, what is the likely bottleneck (effect size, noise, or N)?
2. What is **one** design change to increase power?
 - Case A: outcome is “monthly income”; small expected effect; N is limited.
 - Case B: outcome is a low-variance administrative count; moderate effect; N is moderate.

Assumptions

- A power calculation maps assumptions $\Rightarrow$ probability of detection.
- It's a *design tool* to tell us what to do *under certain assumptions*—
- If the assumptions are wrong, the power calculation can be very misleading.

Minimal detectable effect (MDE)

- Fix α and a target power (e.g., $\alpha = 0.05$, power = 0.80).
- Given N and noise, compute the smallest effect you can reliably detect.
- Alternative reframing: **what can this study rule out?**

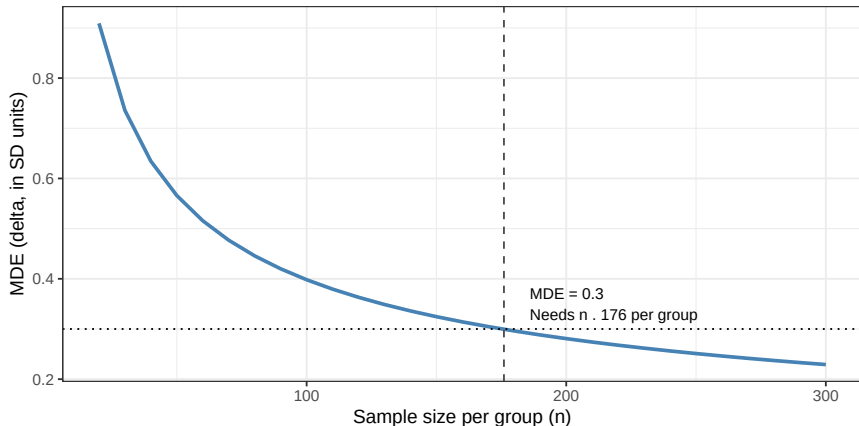
```
alpha <- 0.05
power_target <- 0.80
n_per_group <- 50
sd_y <- 1
mde <- power.t.test(n = n_per_group, power = power_target, sd = sd_y,
                    sig.level = alpha, type = "two.sample",
                    alternative = "two.sided")$delta
cat(sprintf(
  "alpha=%.2f, power=%.2f, n per group=%d (N=%d), sd=%.1f => MDE = %.2f SD\n",
  alpha, power_target, n_per_group, 2 * n_per_group, sd_y, mde
))

## alpha=0.05, power=0.80, n per group=50 (N=100), sd=1.0 => MDE = 0.57 SD
```

MDE decreases as n increases

With fixed noise + alpha, bigger samples detect smaller effects

alpha = 0.05; target power = 0.80; sd = 1



When to believe your power analysis

- **Effect size inputs** are justified.
- **Noise inputs** are justified (pilot data, benchmarks, or conservative assumptions).
- The calculation matches the **design and analysis** you will actually run.

Common failure modes

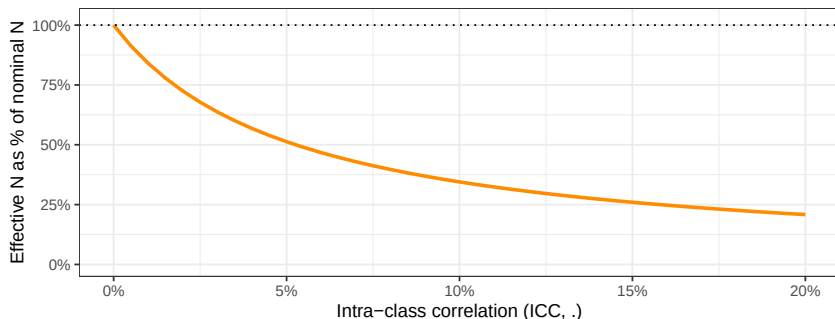
- Using an optimistic effect size from prior literature (especially small studies).
- Ignoring multiple testing or multiple outcomes.
- Ignoring clustering / dependence.

Clustering: the “effective N ” can be much smaller

- If outcomes are correlated within clusters (schools, villages, classrooms), you have less information than N suggests.
- Approximation: design effect $DEFF = 1 + (m - 1)\rho$ where m is cluster size and ρ is the ICC.
- Effective sample size (roughly) $N_{\text{eff}} \approx N / DEFF$.

Even small ICCs can substantially reduce effective sample size

Example cluster size $m = 20$; $N_{\text{eff}} / N = 1 / (1 + (m-1)\rho)$



(Optional) Multiple tests and power

- If you adjust α for multiple tests, power goes down (unless N goes up).
- Simple example: Bonferroni correction lowers the per-test α .
- This is one reason preregistration and a clear analysis plan matter.

Why simulate power?

- Closed-form formulas rely on assumptions that can be wrong.
- Simulation lets you match:
 - your design (assignment ratios, clustering, attrition)
 - your estimator and analysis plan

Simulation recipe

1. Assume an effect size, noise, and design.
2. Generate data from those assumptions.
3. Run the analysis you plan to run.
4. Repeat many times; power $\approx$ fraction of rejections.

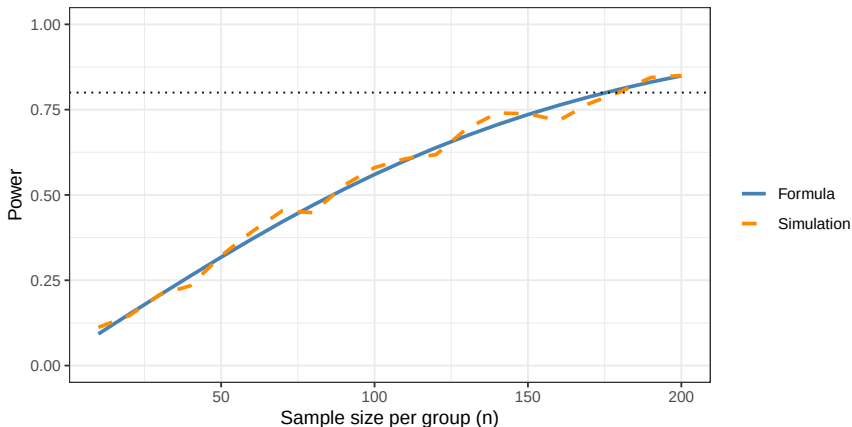
What simulation buys you

- Handles covariates and complicated estimators.
- Handles noncompliance and attrition (if modeled).
- Handles nonlinear outcomes (binary, counts, etc.).
- Helps you sanity-check your design choices.

Power curve in a two-group design

Power increases with n (holding effect size and noise fixed)

Two-sample t-test; $\alpha = 0.05$; $\delta = 0.3$; $sd = 1$; $sims = 500$



- This is power *under the assumptions used to generate the data.*

Sample size justification (Lakens)

Lakens (2022)

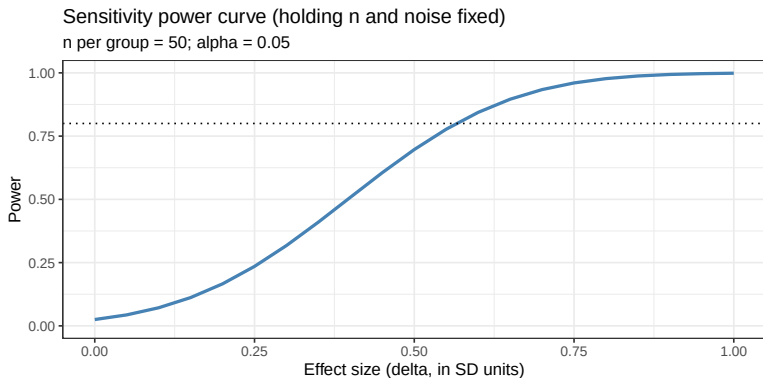
- “Power analysis” is *one* way to justify sample size.
- Other legitimate justifications include:
 - measuring (almost) the full population
 - resource constraints (with transparency)
 - desired accuracy / precision (CI width)
 - heuristics (weaker)
 - explicitly acknowledging no clear justification

What counts as an “interesting” effect?

- Smallest effect size of interest / minimal detectable effect (MDE)
- Expected effect size (with careful justification)
- Confidence interval width (accuracy)
- Sensitivity power analysis over a range of effects
- Distribution of effects in the research area

Sensitivity power vs. “powering for the expected”

- “Powered for expected effect” can be fragile if the expected effect is optimistic.
- Sensitivity power asks: across a range of effects, what can we detect?



Guided example: design + assumptions

- Research question: does a message/frame change support for a policy? Offer-Westort et al. (2026)
- Outcome (Y): standardized support (mean 0, SD 1).
- Estimand: difference in means, $E[Y \mid T = 1] - E[Y \mid T = 0]$.
- MDE: 0.30 SD (small but meaningful).
- Target: $\alpha = 0.05$, power = 0.80, 50/50 assignment.
- Class questions: is this feasible? If not, what lever would you pull first?

Guided example: compute + justify

```
## n per group (80% power, MDE=0.30): 176 (N=352)  
## If n=100, power for MDE: 0.56  
## If n=100, MDE (80% power): 0.40
```

- If the feasible n is too small for your MDE, either change the design (reduce noise) or change the inferential goal (report range of MDEs/sensitivity).

What you will preregister (Week 9)

- Outcome(s) and estimand(s)
- MDE
- Analysis plan (including multiple testing decisions)
- Planned sample size and the justification approach

In one sentence: what would you change to increase power in your own design?

References I

Coppock, A. (2022). 10 things to know about statistical power. EGAP.

Lakens, D. (2022). Sample size justification. *Collabra: Psychology*, 8(1):33267.

Offer-Westort, M., Gruen, W., Herron, C., Hyatt, K., Buford, M., Davis, K., Fonseca, D., Gurevich, M., Huang, T., Jerez, R., et al. (2026). Which frame fits? policy learning with framing for climate change policy attitudes. *Research & Politics*, 13(1):20531680251414927.